

EXPLORATORY DATA ANALYSIS OF YOUTUBE TRENDING VIDEOS DATASET

Content Trends, Viewership
Dynamics & User Engagement



TOOLS USED

Python, Pandas, Matplotlib, Seaborn & Plotly

PRESENTED BY

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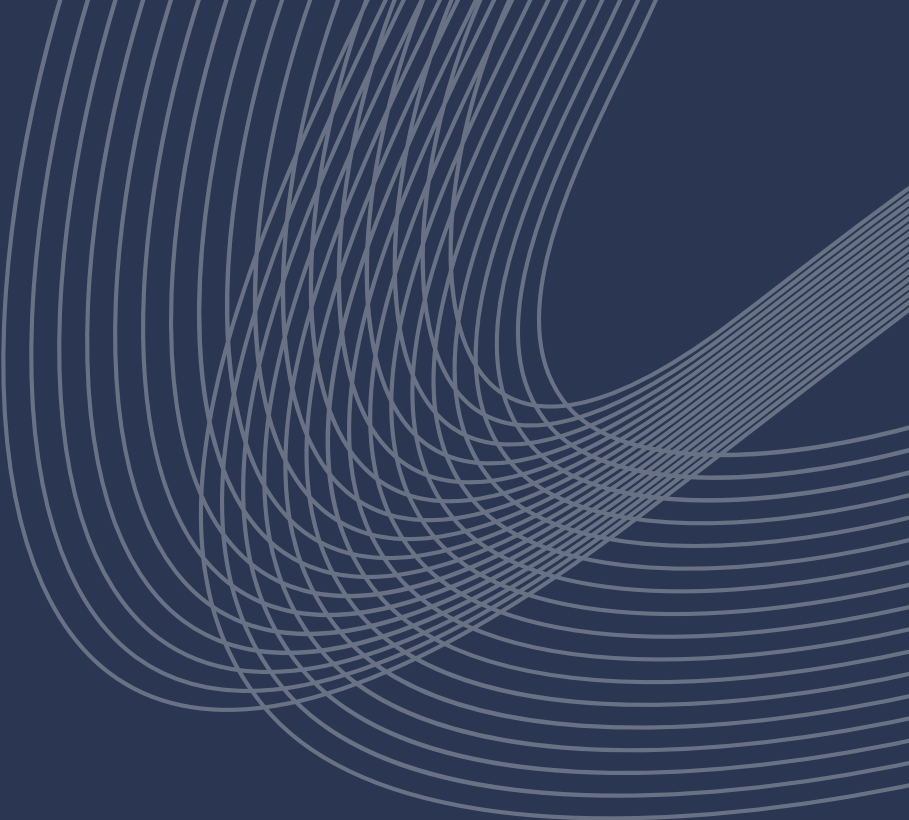


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PROBLEM STATEMENT & OBJECTIVE

Problem Statement:

The YouTube platform hosts millions of videos across diverse categories. With massive variations in views, likes, and engagement, it becomes challenging to understand content trends, audience preferences, and what drives video success.

Objective:

To explore the YouTube trending videos dataset and uncover patterns, insights, and trends that help creators and businesses make data driven decisions.



EDA WORKFLOW

For this analysis, a structured workflow was followed, involving data collection, understanding, cleaning, exploration, and summarization of insights to allow a clear understanding of the dataset and its trends.

01

Data Collection

- Gathered the Google Play Store dataset from Kaggle

02

Data Understanding & Anomaly Detection

- Looked at data distributions
- Found missing values, outliers, and unusual patterns

03

Data Cleaning & Treatment

- Fixed missing or incorrect values
- Standardized formats for consistency

04

Exploratory Analysis

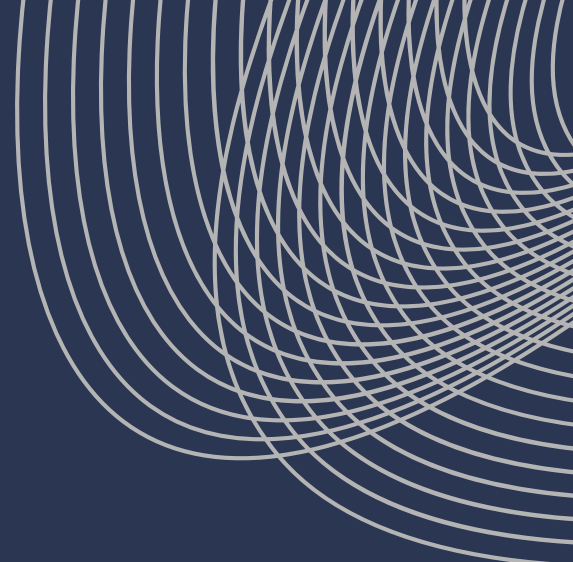
- Univariate Analysis: e.g., Ratings, Installs, Prices
- Bivariate Analysis: Rating vs Reviews
- Multivariate Analysis: Reviews vs Rating vs Installs

05

Insights & Reporting

- Summarized patterns and trends
- Highlighted key findings for developers and businesses

KEY QUESTIONS EXPLORED



- Which categories dominate the Play Store, and where is competition highest?
- What is the distribution of Free vs Paid apps across categories and ratings?
- Who are the target audiences of apps (content rating distribution)?
- How do app sizes vary, and what is the optimal size range?
- How do app ratings behave across categories, review volumes, and time?
- What monetization models work best by category and audience?



DATA OVERVIEW

The dataset provides key information about YouTube trending videos, including views, likes, comments, category, publish date, and trending date. Additional details such as channel information and country capture content performance, audience engagement and regional trends.

Data Source: Kaggle

Dataset Size

161470

Records

18

Features

App Diversity

56905

Videos covered

4

Countries

18

Video Categories



DATA OVERVIEW

Below is a detailed description of the feature set:

Dataset Features	Type	Feature Description
Index	Numerical (Discrete)	Dataset row number
Video_id	String	Unique YouTube video ID
Trending_date	Date	Date the video trended on YouTube
Title	String	Video title
Channel_title	String	Publishing channel name
Category_id	Numerical (Discrete)	YouTube category ID
Publish_date	Date	Video publication date
Time_frame	Categorical	Price of the app (0 indicates a free app)
Published_day_of_week	Date	Day of the week published
Publish_country	Categorical	Country of publication
Tags	String	The tags or keywords associated with the video.
Views	Categorical	The number of views received by a particular video
Likes	Numerical (Discrete)	The number of likes received by a particular video
Dislikes	Numerical (Discrete)	The number of dislikes received by a particular video
Comment_count	Numerical (Discrete)	The number of comments received by a particular video
Comments_disabled	Boolean	Whether comments are disabled
Ratings_disabled	Boolean	Whether ratings are disabled
Video_error_or_removed	Boolean	Whether the video is unavailable, removed, or errored

DATA QUALITY CHALLENGES & ANOMALIES

Few inconsistencies were found in the dataset, which could have affected the analysis if left unaddressed.

DATA ANOMALIES

- **Columns requiring cleaning or type conversion:** trending_date, publish_date
- **Columns with notable anomalies or invalid entries:**
 - tags (very long strings, |-separated values)
 - publish_country (may require standardization with short forms)



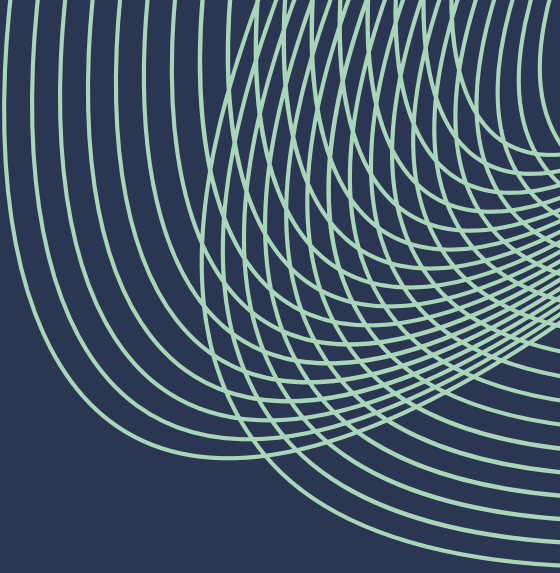
DATA CLEANING & TREATMENT

Inconsistent and missing values were addressed, and key features were cleaned and standardized for analysis.

DATA CLEANING SUMMARY

- Features such as **trending_date**, **publish_date** were standardized with respect to data type conversion.
- Short forms the **publish_country** feature were standardized.
- The feature **tags** was transformed into a clean, standardized strings by splitting, normalizing, deduplicating and removing irrelevant tags.
- New features **days_to_trend** & **category_name** were created from **trending_date**, **publish_date** & **category_id** respectively.
- **index** and **video_id** was removed as it is a technical detail with limited business relevance.





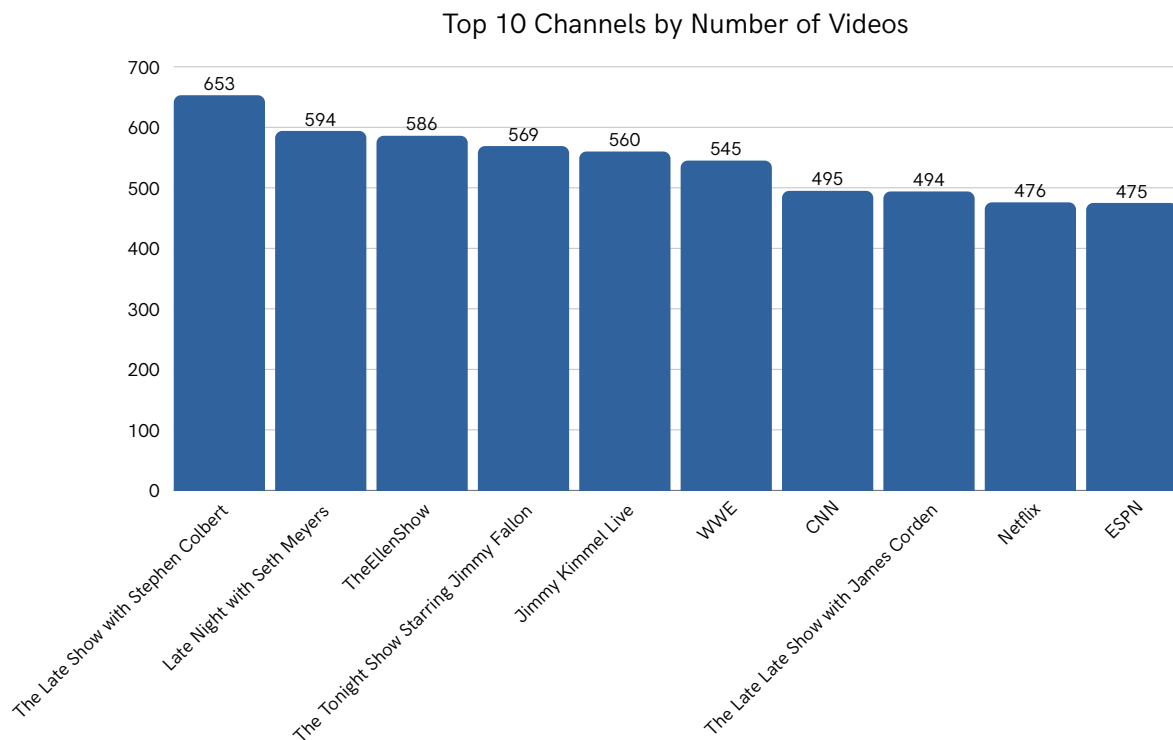
INSIGHTS



Trending Exposure Is Concentrated Among a Few Dominant Channels

5,447

Videos from Top 10 Channels



Key observations

- A small group of channels (late-night shows, major media brands) dominate the number of videos, while most creators appear far less frequently.

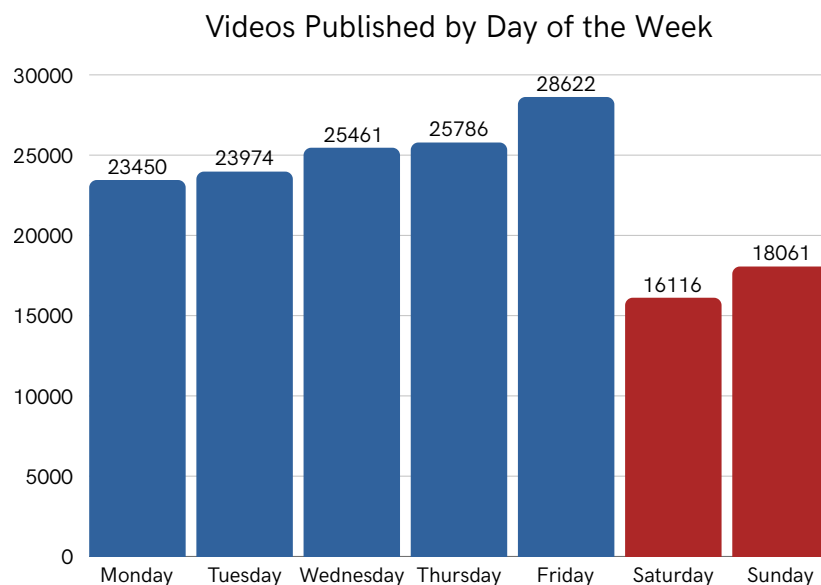
Business Insights

- Visibility is heavily concentrated among established brands.
- New or smaller channels need niche content or viral strategies to compete effectively.

Video Publishing Peaks on Weekdays and Drops on Weekends

23,067

Average Weekly Upload Volume



Key observations

- Video publishing peaks on weekdays, especially Friday and drops significantly over the weekend.

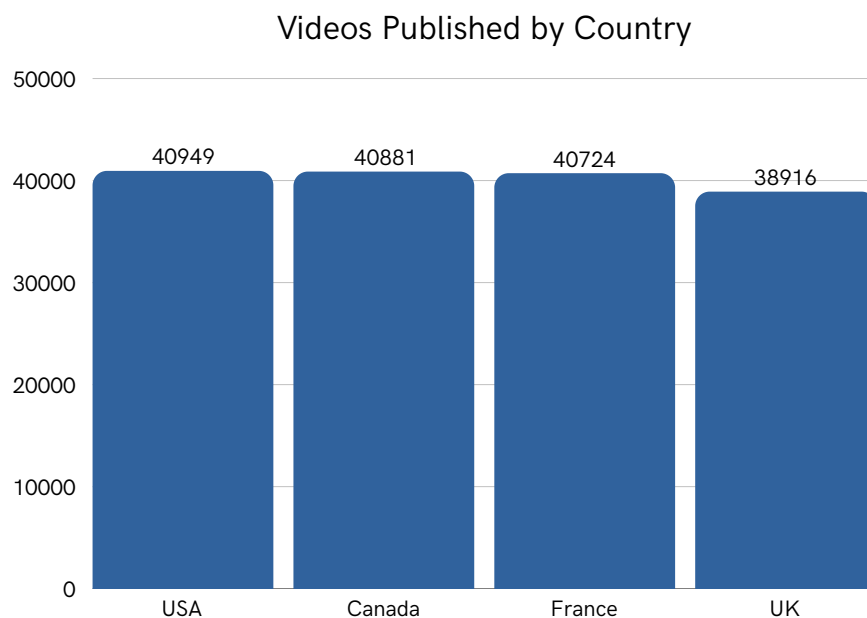
Business Insights

- Uploading during weekdays, particularly **Thursday-Friday** can improve visibility and engagement, while weekend uploads face lower competition but reduced overall activity.

Content Creation Is Evenly Distributed Across Major Countries

1,61,470

Total Videos from 4 Major Countries



Key observations

- Video publishing volume is fairly balanced across countries, with the USA, Canada and France contributing slightly more content than the UK.

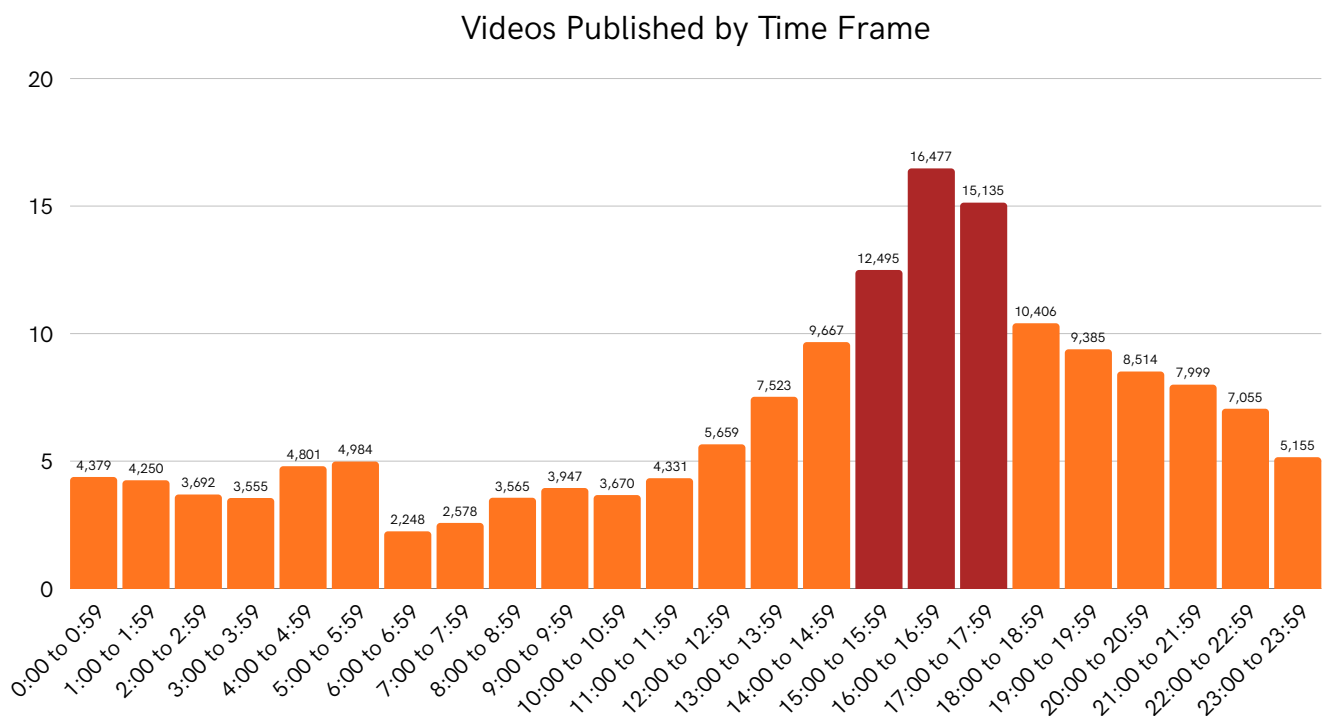
Business Insights

- The near-uniform distribution suggests global participation in content creation, so region-specific optimization (timing, language, trends) may be more effective than focusing on a single dominant market.

Most Videos Are Uploaded Between 3 PM and 6 PM

1,61,470

Total Videos from 4 Major Countries



Key observations

- Video uploads are heavily concentrated between 3 PM and 6 PM, with a clear peak around 4–5 PM.
- Early morning hours show the lowest publishing activity, indicating reduced creator presence.

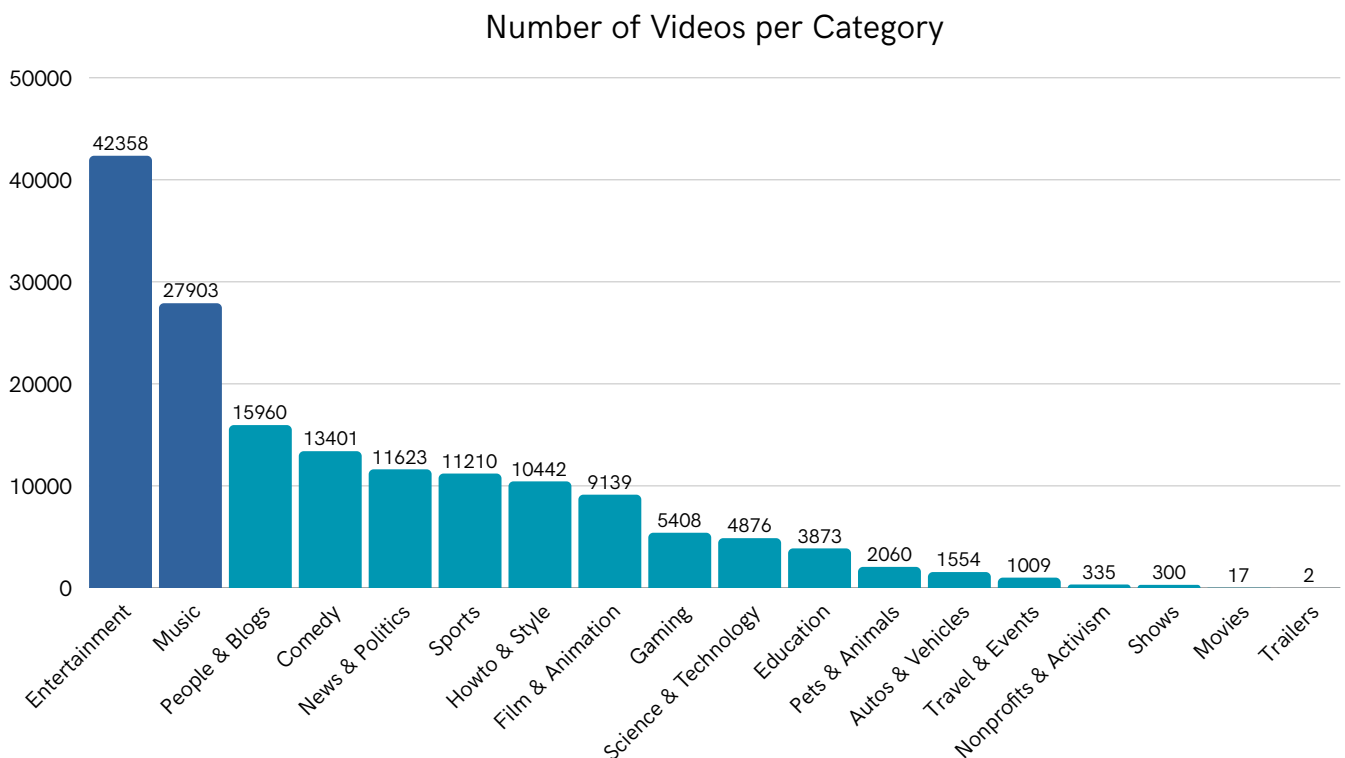
Business Insights

- Creators usually publish during peak audience hours, creating heavy competition, while late-night and early-morning windows may offer better visibility due to lower content saturation.

A Few Content Categories Dominate YouTube Upload Volume

43.51%

Of the Videos are in Music & Entertainment category



Key observations

- Entertainment and Music alone account for a disproportionately large share of total video uploads.
- Most other categories form a long tail with significantly lower content volume.

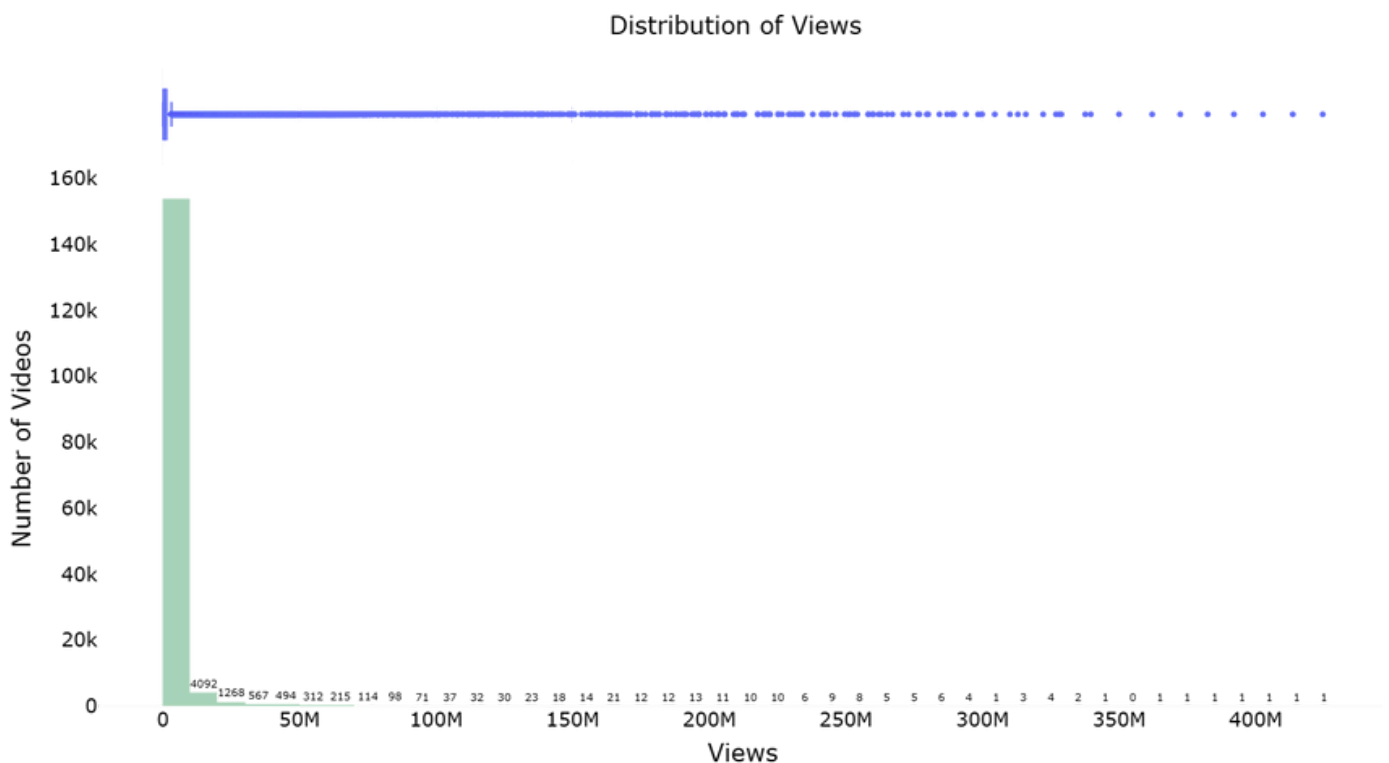
Business Insights

- High-volume categories face severe competition, making discoverability harder for new creators.
- Niche and under-served categories (eg. Education, Science & Technology) present opportunities for differentiation and targeted audience growth.

Most Videos Trend Quickly, Late Breakouts Are Rare

3 Days

Median Time Taken to Trend



Key observations

- Video views are extremely right-skewed, with the vast majority of videos receiving relatively low views and a small fraction achieving very high view counts.

Business Insights

- Success is driven by rare viral hits, so relying on a few breakout videos is risky
- Consistent publishing and optimization are key to improving average performance over time.